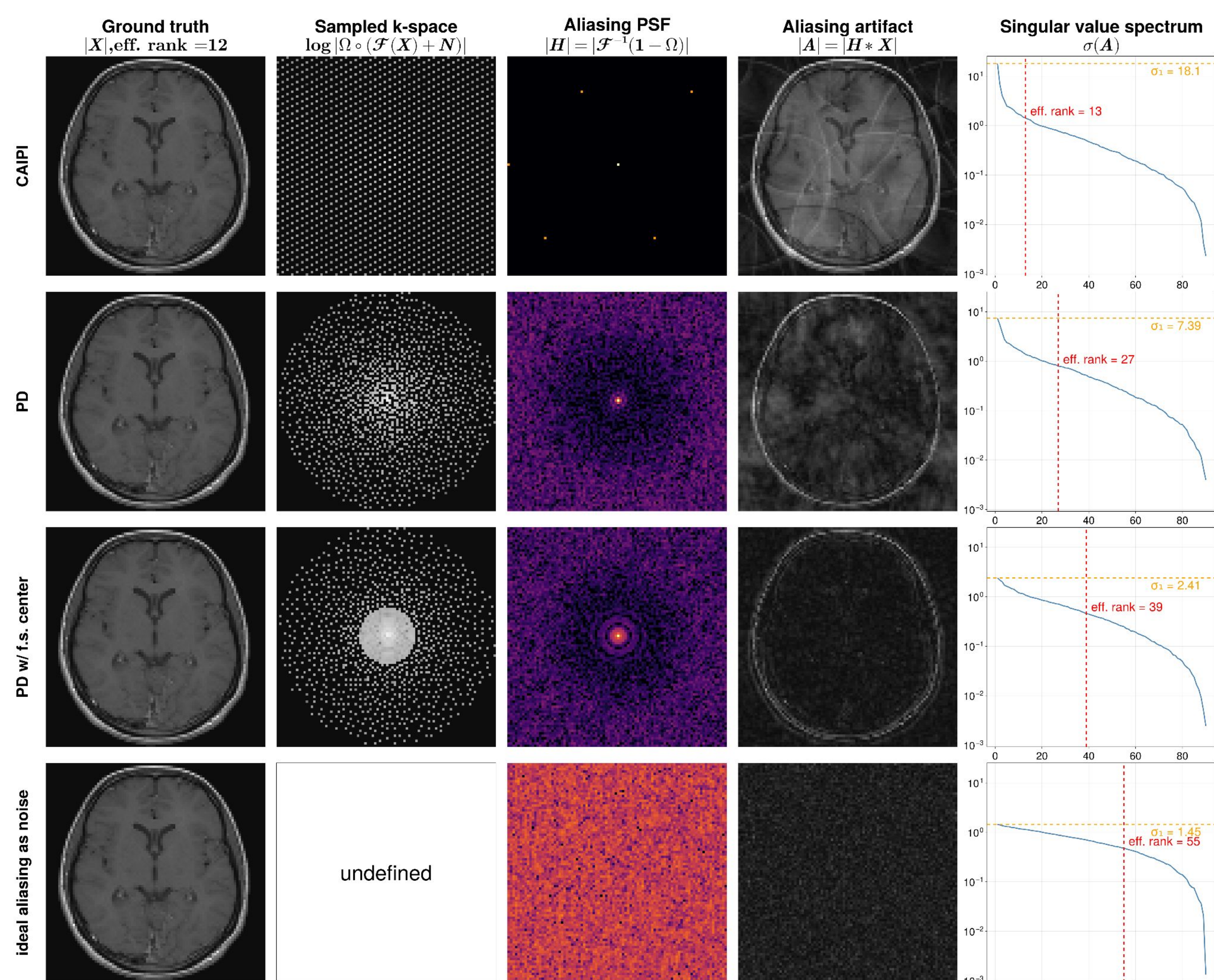


Introduction: random sampling strategies for functional MRI (fMRI)

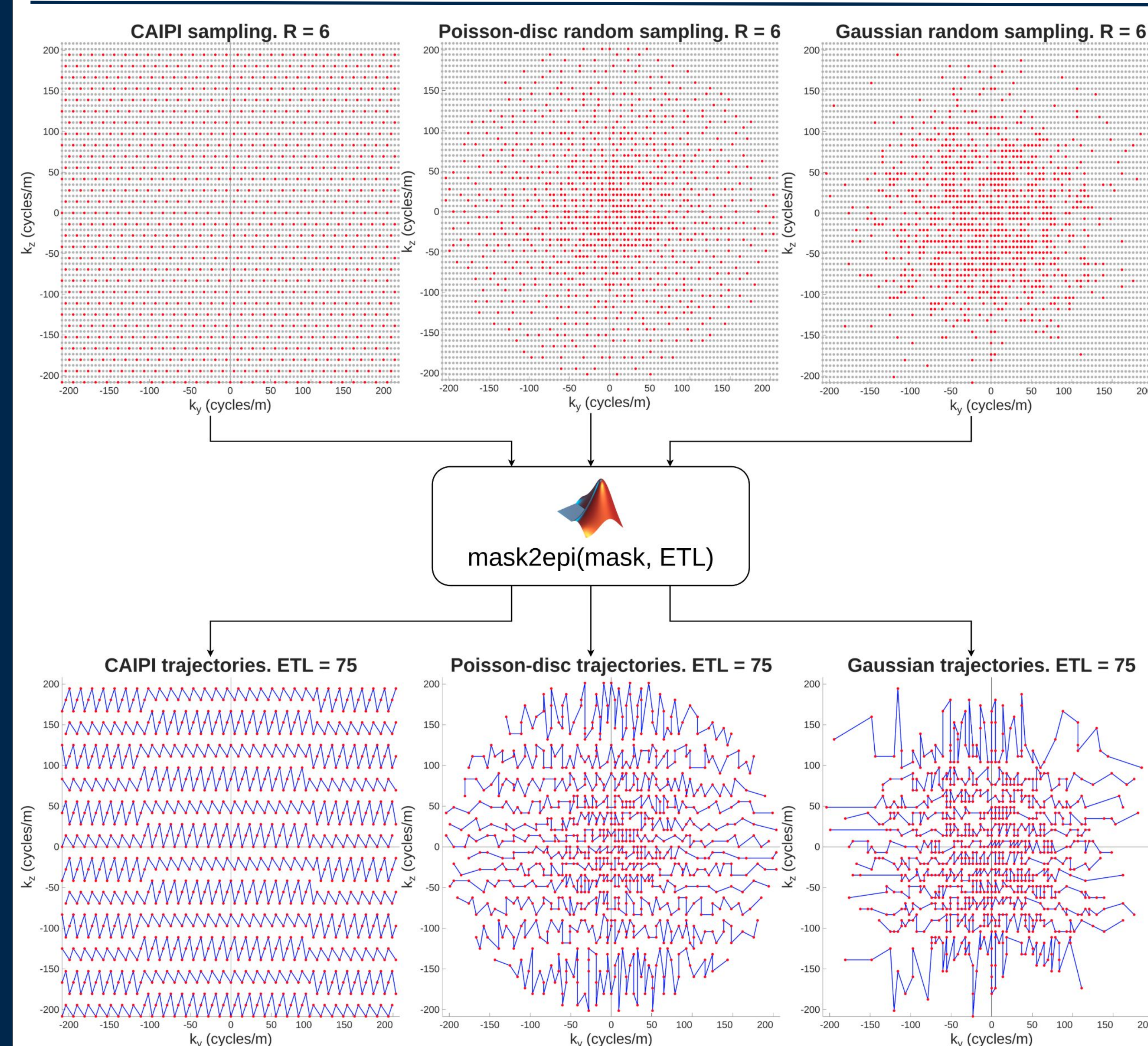
- Compressed sensing [1] k-space sampling strategies, e.g., Poisson-disc (PD) are routinely used in dynamic MRI [2]
- Unclear how to apply CS sampling principles to BOLD fMRI, and how to efficiently implement random ky-kz undersampling in 3D EPI
- Major implementation challenge from a pulse sequence programming perspective
- We provide a framework for generating scanner hardware compliant 3D-EPI sequences from arbitrary 2D Cartesian sampling patterns in ky-kz
- By decoupling the sequence programming and sampling design problems, researchers can focus on the latter and rapidly test on real scans

Background: the 2D sampling design problem



- Eff. rank = # of singular values accounting for 95% of signal energy
- Our ArbEPI toolbox supports arbitrary ky-kz sampling patterns, such as the three examples shown here
- Regular (CAIPI) sampling (1st row) results in coherent aliases, with structure indicated by a low effective rank and large spectral norm
- Random Poisson-Disc (PD) sampling (2nd row) results in incoherent aliases, indicated by higher effective rank and lower spectral norm, which is further improved with a fully sampled core (3rd row)
- Bottom row shows complex Gaussian noise (with equivalent total energy as the previous aliasing artifact) as the ideal aliasing artifact, which is mathematically unattainable with a binary sampling mask
- Nonetheless, we want a sampling mask that drives aliasing as close to noise as possible; ArbEPI aims to accelerate the testing of that mask on a real scanner

Methods: creating efficient 3D-EPI sequences

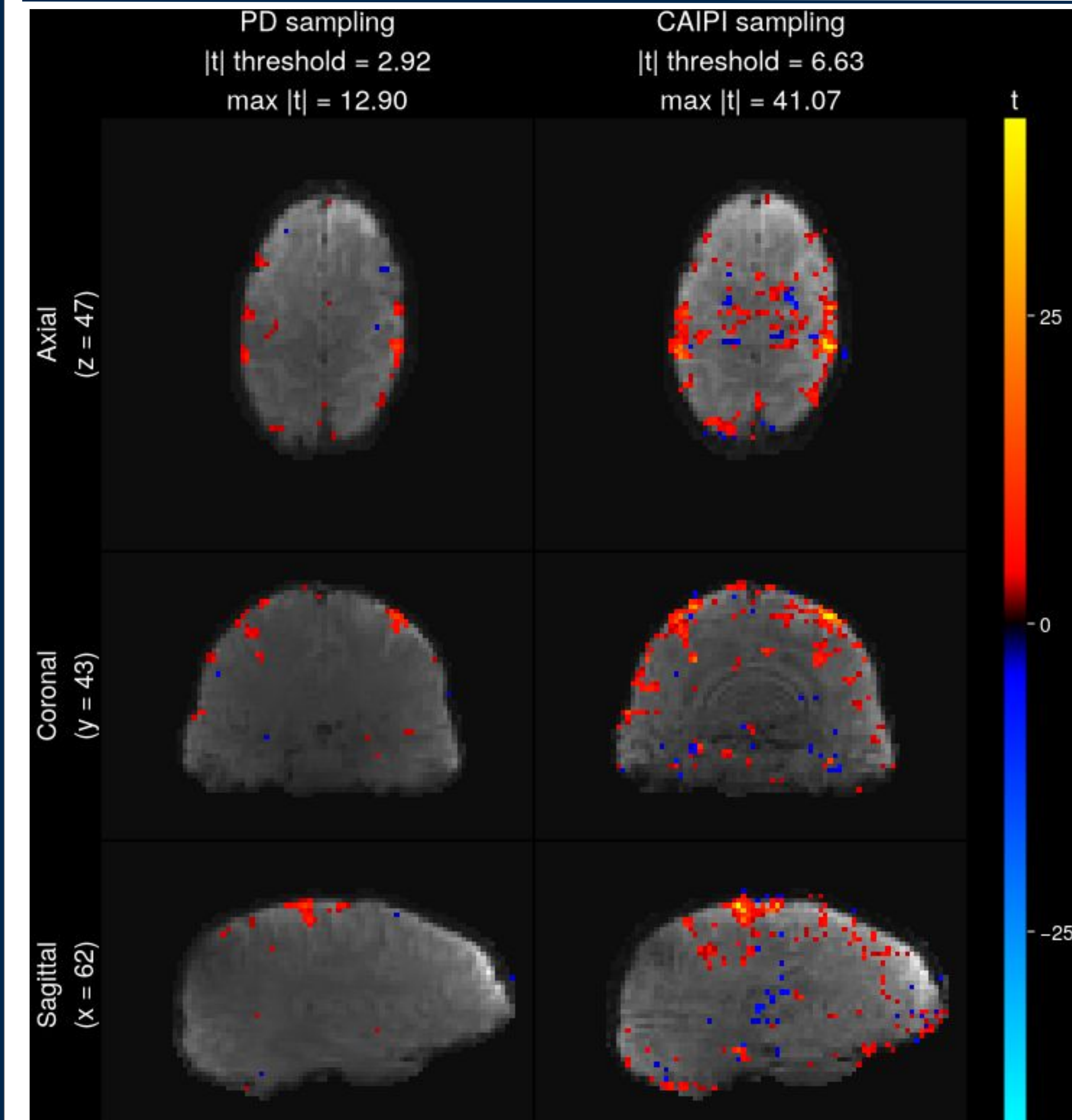


- The core function is mask2epi, which generates a 3D-EPI trajectory (blue lines) from any 2D sampling mask (red dots) and prescribed echo train length (ETL), called once per frame to generate temporally varying/independent sampling patterns
- First, samples are partitioned into shots by selecting samples with a center-out approach. This results in approximately the same number of samples about $k_y = 0$, thus allowing control of the effective TE to achieve BOLD contrast for fMRI
- Then, samples within each shot are strung together via the shortest path to minimize gradient blip requirements. k_y is constrained to be nondecreasing to keep the intrashot phase accumulation linear, so that they are compatible with classic EPI phase correction methods
- Sequence execution speed is limited by the largest gradient blip, which corresponds to the largest jump in k-space. In this example, CAIPI is the fastest, followed by PD and Gaussian
- Sequences were programmed using Pulseseq [3], a vendor-neutral pulse sequence programming platform, with careful block boundary shifting to minimize scanner waveform memory footprint
- For fMRI demonstration, we programmed CAIPI and PD sampled 3D EPI sequences, each being 5 minutes long (375 frames)

References

- [1] Lustig, M., Donoho, D., & Pauly, J. M. (2007). Sparse MRI: The application of compressed sensing for rapid MR imaging. *MRM*, 58(6), 1182-1195.
- [2] Otazo, R., Candes, E., & Sodickson, D. K. (2015). Low-rank plus sparse matrix decomposition for accelerated dynamic MRI with separation of background and dynamic components. *MRM*, 73(3), 1125-1136.
- [3] Layton, K. J., Kroboth, S., Jia, F., Littin, S., Yu, H., Leupold, J., ... & Zaitsev, M. (2017). Pulseseq: a rapid and hardware-independent pulse sequence prototyping framework. *MRM*, 77(4), 1544-1552.
- [4] Uecker, M., Ong, F., Tamir, J. I., Bahri, D., Virtue, P., Cheng, J. Y., ... & Lustig, M. (2015, May). Berkeley advanced reconstruction toolbox. In *Proc. Intl. Soc. Mag. Reson. Med* (Vol. 23, No. 2486, p. 9).

Results: demonstration in a finger tapping experiment



- Representative fMRI results using 3D-EPI sequences designed with the proposed toolbox. Left: Temporally varying PD sampling. Right: conventional CAIPI sampling. Reconstructed using L1-wavelet + total variation regularized SENSE via the BART toolbox [4]
- Both PD and CAIPI sampled images reveal motor activation as expected from a finger tapping task
- PD images have less aliasing, CAIPI images have higher overall t-scores, but the lower t-scores for PD are likely due to time-varying sampling and can be restored with spatiotemporal reconstruction

Conclusions

- ArbEPI is a framework that allows researchers to rapidly test 2D sampling designs for 3D dynamic MRI with minimal pulse sequence programming experience requirements
- We demonstrate the toolbox's feasibility by implementing 3D-EPI sequences with standard CAIPI and random PD sampling, both of which produced motor activation in a finger tapping study
- Many open questions remain regarding the optimal sampling strategy for 3D dynamic MRI; we invite you to use our toolbox to investigate novel sampling strategies (<https://github.com/rextlfung/ArbEPI>)
- Future directions:
 - Shuffle shot ordering to randomize multishot artifacts
 - Enable trajectories to traverse in any direction, which may improve k-space sampling efficiency by having e.g. radial-like trajectories
 - Generalize trajectories to spokes that can begin anywhere to enable, e.g., spiral-like trajectories. Doing so increases the range of possible TEs e.g. zero TE imaging